SpectraSense

Smart Conveyor Classification System

IoT · Sensor Fusion · Computer Vision · YOLO AI · Real-Time Detection

Technical Project Documentation

Integrating IoT Sensor Technology with AI-Powered Object Recognition

1. Introduction

SpectraSense is a smart conveyor-based classification system designed to detect, identify, and sort objects automatically using a combination of IoT sensor technologies and AI-powered computer vision. The system integrates an inductive proximity sensor, a Hall Effect sensor, and a real-time YOLO object detection model to classify objects both by material composition and by visual identity as they travel along a moving conveyor belt.

The project is divided into two tightly integrated subsystems: the IoT layer, which handles physical sensing and actuation, and the AI layer, which performs visual inference. Together they form a unified pipeline capable of delivering richer classification decisions than either subsystem could achieve alone.

Key Technologies

Technology	Role
LJ18A3-8-Z/BX Inductive Sensor	Metal vs. non-metal detection (6–36V DC, electromagnetic induction)
Hall Effect Sensor	Ferromagnetic vs. non-ferromagnetic metal discrimination
Conveyor Belt & Drive Motors	Mechanical transport and object positioning
YOLOv8 + OpenCV	Real-time object recognition and visual classification
Microcontroller (MCU)	Sensor reading, data acquisition, and actuation control

2. Problem Statement

Traditional automated sorting systems rely on a single sensing modality, which significantly limits their ability to accurately distinguish between different material types — particularly when differentiating between ferromagnetic and non-ferromagnetic metals. A standard inductive sensor can confirm the presence of metal, but cannot determine whether that metal is iron, steel, copper, or aluminum.

Furthermore, most existing industrial sorting systems do not integrate visual recognition capabilities. This prevents them from identifying what an object actually is — its form, shape, and category — alongside its material properties. Knowing that an object is metallic is insufficient if the system cannot distinguish between a key, a coin, and a bolt.

This lack of multi-modal integration creates a significant technological gap in high-precision recycling, manufacturing, and logistics applications, where both accurate material detection and object-level classification are essential for reliable and efficient sorting.

Core Limitations of Existing Systems

- Single-sensor architectures cannot sub-classify metal types without additional sensing.
- Absence of computer vision prevents object identity recognition alongside material detection.
- No integration layer to fuse sensor and visual data into a unified classification decision.
- Limited applicability to high-precision industrial sorting requiring both material and object-level intelligence.

3. Project Goals & Objectives

SpectraSense is designed to address the limitations described above through five core objectives:

Objective	Description
Metal Detection	Detect whether an object is metallic or non-metallic using the LJ18A3-8-Z/BX inductive proximity sensor.
Material Classification	Distinguish between ferromagnetic metals (iron, steel) and non-ferromagnetic metals (copper, aluminum) using Hall Effect sensor readings.
Object Recognition	Identify and classify objects by visual appearance (bottles, keys, boxes, and other daily-carry items) using a YOLO-based computer vision model achieving ≥89% accuracy per class.
System Integration	Combine sensor data and visual inference outputs into a unified decision layer for automated, real-time sorting.
Real-Time Performance	Achieve continuous real-time classification suitable for a conveyor-belt industrial environment without manual intervention.

4. Architecture

SpectraSense is structured as a three-layer architecture. Data originates from physical sensors and a camera, flows through parallel IoT and AI processing pipelines, and is fused at the Decision & Control layer to produce a final sorting command.

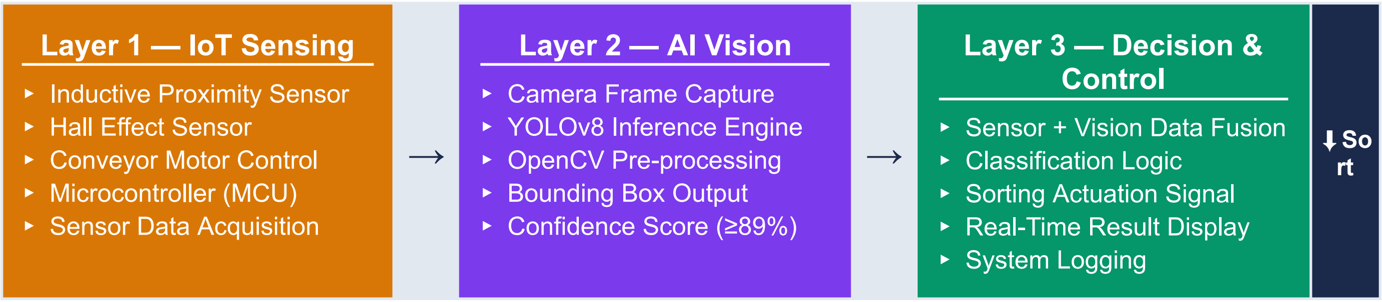

Figure 1: SpectraSense three-layer system architecture and data flow.

4.1 IoT Sensing Layer

As an object travels along the conveyor belt, the inductive proximity sensor detects whether it is metallic or non-metallic. If metal is confirmed, the Hall Effect sensor measures the object's magnetic field response to discriminate between ferromagnetic and non-ferromagnetic metals. All sensor readings are acquired and processed by the microcontroller.

Sensor Output	Material Classification
Proximity: No Detection	Non-metallic — Plastic, Glass, Rubber, etc.
Proximity: Metal + Hall: Strong	Ferromagnetic metal — Iron, Steel
Proximity: Metal + Hall: Weak	Non-ferromagnetic metal — Copper, Aluminum

4.2 AI Vision Layer

A camera mounted above the conveyor continuously captures frames. Each frame is processed by a trained YOLO model that outputs a class label, bounding box coordinates, and a confidence score for the detected object. Object classes include bottles, keys, boxes, and other commonly carried items. A minimum accuracy threshold of 89% per class is enforced before any label is considered production-ready.

4.3 Decision & Control Layer — Data Fusion

The fusion layer combines outputs from both subsystems using timestamp alignment to pair each sensor reading with the corresponding YOLO inference result. The final classification is richer than either subsystem alone: a 'copper key' is distinguishable from an 'iron key' despite identical visual appearance, and a 'plastic bottle' is distinguishable from a 'metal bottle' despite similar shape.

Proximity Sensor	Hall Effect	YOLO Vision	Final Classification
No Metal	—	Bottle	Non-metal Bottle
Metal Detected	Strong (Ferromagnetic)	Key	Ferromagnetic Key — Iron/Steel
Metal Detected	Weak (Non-ferromagnetic)	Key	Non-ferromagnetic Key — Copper
Metal Detected	Strong (Ferromagnetic)	Box	Ferromagnetic Metal Box

Table 1: Fusion logic — combined sensor and vision outputs for final classification.

5. Project Phases

The project is structured across the following phases, covering both the IoT and AI subsystems:

5.1 IoT Phases

- Research, select, and purchase the required electronic components based on system requirements and design specifications.
- Research sensors for metal detection and material type identification.
- Develop the mechanical design and assemble the system, including installation of the conveyor belt and drive motors.
- Integrate and connect electronic components with the hardware platform.
- Write and develop the firmware code for sensor acquisition and system control logic.
- Test the code on hardware to verify functionality and system performance.
- Analyze Hall Effect sensor readings to calibrate thresholds for distinguishing ferromagnetic and non-ferromagnetic materials.
- Integrate the hardware system with the AI/computer vision pipeline.

5.2 AI Phases

Phase 1 — Data Collection & Preparation

- Define and select object classes relevant to the conveyor classification use case.
- Collect labeled image data from multiple sources: Roboflow, Kaggle, and open-source repositories.
- Filter, clean, and manually annotate all images; organize into a shared Roboflow workspace with version control.

Phase 2 — Model Training & Optimization

- Select the appropriate YOLO architecture version and configure training parameters (image size, batch size, epochs, confidence thresholds).
- Train the model and monitor convergence; evaluate per-class accuracy using a strictly separated test set.
- Iterate on underperforming classes using data augmentation, class rebalancing, or architecture adjustment until all labels reach ≥89% accuracy.

Phase 3 — Final Testing & Deployment

- Perform a full dataset audit before the final training run: verify splits, consistency, and label quality.
- Integrate the trained YOLO model with OpenCV for live camera inference.
- Validate the system in real-world conditions: varied lighting, motion blur, partial occlusion, and diverse backgrounds.

6. Finished Phases

6.1 IoT — Completed

- ✅ Electronic components researched, selected, and purchased based on system requirements and design specifications.
- ✅ Proximity sensor and Hall Effect sensor research completed for material type identification.
- ✅ Mechanical design finalized; conveyor belt and drive motors assembled and installed.
- ✅ Electronic components integrated and wired with the hardware platform.
- ✅ Firmware code written and developed for sensor acquisition and control logic.

6.2 AI — Completed

- ✅ Object classes defined and selected based on real-world relevance and visual distinctiveness.
- ✅ Training dataset collected from Roboflow and Kaggle; filtered, cleaned, and manually annotated.
- ✅ Shared Roboflow workspace established for version-controlled dataset management.
- ✅ YOLO model architecture selected and training parameters configured.
- ✅ Model trained, evaluated per class, and iterated until all labels reached ≥89% detection accuracy.
- ✅ Final dataset audit completed; definitive dataset version locked in Roboflow.
- ✅ YOLO model integrated with OpenCV; live camera inference validated in real-world conditions.

6.3 Integration Status

Milestone	Status	Team
Hardware assembly & sensor installation	✅ Complete	IoT
MCU firmware for sensor acquisition	✅ Complete	IoT
YOLO model training (≥89% per class)	✅ Complete	AI
OpenCV live inference integration	✅ Complete	AI
Hardware code validation & testing	🔄 In Progress	IoT
Hall Effect threshold calibration	🔄 In Progress	IoT
Sensor + vision data fusion layer	🔜 Planned	Both
End-to-end system testing	🔜 Planned	Both

7. Next Steps

7.1 IoT — Immediate Next Steps

- Complete code validation on hardware to verify sensor acquisition and control logic under real operating conditions.
- Analyze Hall Effect sensor readings to define accurate threshold boundaries between ferromagnetic and non-ferromagnetic materials.
- Investigate more advanced sensing components or methods to achieve higher-resolution identification of different metal types.
- Optimize power consumption of the MCU and sensor array for continuous industrial operation.

7.2 AI — Immediate Next Steps

- Expand the object class library by adding new categories (glasses, headphones, umbrellas) through the same rigorous pipeline, each meeting the 89% accuracy threshold.
- Develop a user-facing GUI for public testing, enabling non-technical users to test the system live via their own camera.
- Integrate a structured feedback form after each user session to collect detection accuracy ratings and missed-class reports.

7.3 Integration — Planned

- Build the sensor + vision data fusion layer to combine IoT sensor readings with YOLO inference outputs into a unified classification decision.
- Develop and test the sorting actuation logic based on fused classification results.
- Conduct end-to-end system testing across all object categories and material combinations.

7.4 Future Enhancements

Enhancement	Description
Edge Deployment	Quantize and prune the YOLO model for deployment on Raspberry Pi, mobile, or embedded devices.
Multi-Layer Object Identity	Extend beyond category classification to specific object identity (e.g., coin vs. key within the 'metal' class).
Adaptive Hall Calibration	Implement dynamic threshold adjustment for Hall Effect readings under varying environmental conditions.
Remote Monitoring Dashboard	Add a real-time web dashboard for monitoring classification results, system health, and sensor logs.